

Supplementary material: record boards and construction histories

for *How much can a Scrabble move score? Exact records from one tile to seven*

Rules and certificate guide

Columns A–O run left to right and rows 1–15 top to bottom; H8 is the centre. The first move covers H8. A later move places one to seven tiles in one row or column. Every new maximal string of length at least two must be in the stated dictionary. Premiums apply only when first covered. A seven-tile move earns 50 extra points. **Tile supply and face values.**

Tile	Copies	Value	Tile	Copies	Value
A	9	1	N	6	1
B	2	3	O	8	1
C	2	3	P	2	3
D	4	2	Q	1	10
E	12	1	R	6	1
F	2	4	S	4	1
G	3	2	T	6	1
H	2	4	U	4	1
I	9	1	V	2	4
J	1	8	W	2	4
K	1	5	X	1	8
L	4	1	Y	2	4
M	2	3	Z	1	10
Blank	2	0			

Premium squares.

TW, DW, TL, and DL mean triple word, double word, triple letter, and double letter, respectively.

TW A1, H1, O1, A8, O8, A15, H15, O15.

DW B2, N2, C3, M3, D4, L4, E5, K5, H8, E11, K11, D12, L12, C13, M13, B14, N14.

TL F2, J2, B6, F6, J6, N6, B10, F10, J10, N10, F14, J14.

DL D1, L1, G3, I3, A4, H4, O4, C7, G7, I7, M7, D8, L8, C9, G9, I9, M9, A12, H12, O12, G13, I13, D15, L15.

Dictionary identity.

NWL23 binary SHA-256: 3e74af981fdd974e107283f686da0fe4

b7ec84ad0d825d444330c338c33b91ba.

NWL23 sorted word-list SHA-256: 5254d44586c3af6a8530427b14ef0251

af5aa47906ce32a45e4c229261b1456b.

CSW24 binary SHA-256: 62ca7a84f07429a9976f77a4f74b9491

1aca5d49cce050dce72dd0e032c0566f.

CSW24 sorted word-list SHA-256: 5646ebbca6b1d5e57459ec431ee81aa8

f627e81ab121fe9c7a815221cb727912.

NWL23 has 212,868 exported entries, of which 196,601 have board length 2–15. CSW24 has 299,162 exported entries, of which 280,887 have board length 2–15.

The binary dictionaries are decoded as little-endian 32-bit directed-acyclic-word-graph nodes. A depth-first traversal emits a word precisely at an accepting node; sorting the emitted uppercase strings gives the word lists hashed above.

Machine-readable certificates.

certificates/record_witnesses.json: 158c30e829bf88cd189eff063e333b29

26fd7f48e9e0b3d59377f0caebdac677.

certificates/one_tile_certificate.json: 033a6d7681face564db6c049db51dff1

43bc32476613d3eac72c4afb7dd9d4b3.

certificates/upper_bound_audit.json: c024b76e9414755c9a65dea7291bcfa0

827478c7febac292c1c19e5f0093d6d7.

certificates/reproduction_verified.json: 5747a4d89de6b900b66ce9fc08a2fe8f

1280011b345fcb6fe792041460cf7bd5.

certificates/fresh_run_state.json: 0cab4e8bdc668f654a770282678d8af

b792caee6c0403681b1d254267b86b91.

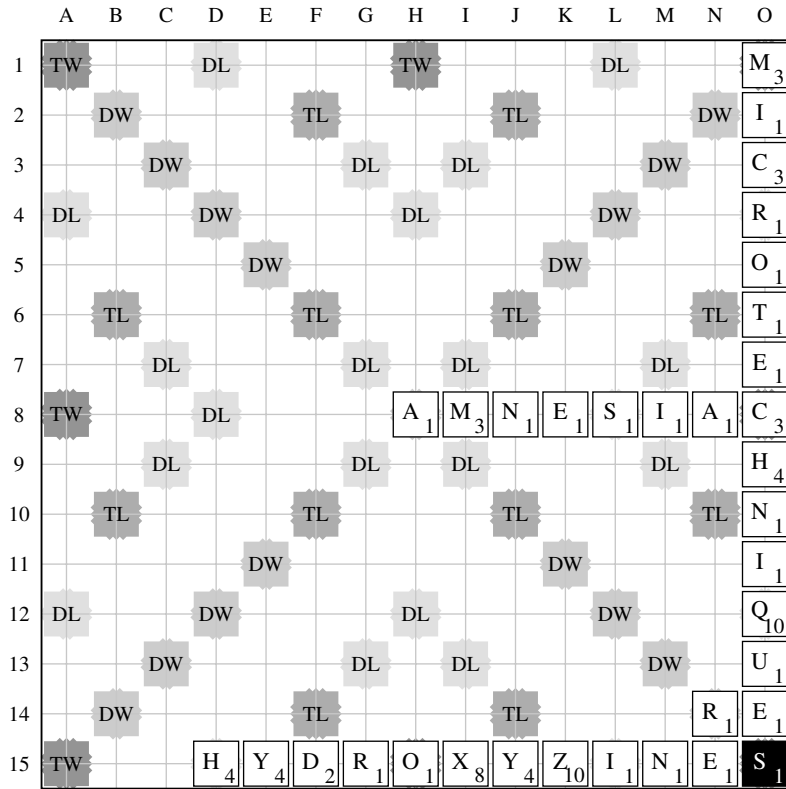
certificates/nwl23_k7_rack_certificate.json: 39cc5f1295097eb5ffb2472a038d579e

47daa2e140ca3818d5bfa125fb2e9691.

The public driver scripts/reproduce_release.py regenerates every upper-bound search and invokes separate forward verifiers for every history.

The following tables list every tile-placement turn; omitted intervening turns are passes, while rack refills and draws remain in the JSON certificates.

NWL23: $k = 1$, 213 points



Rack

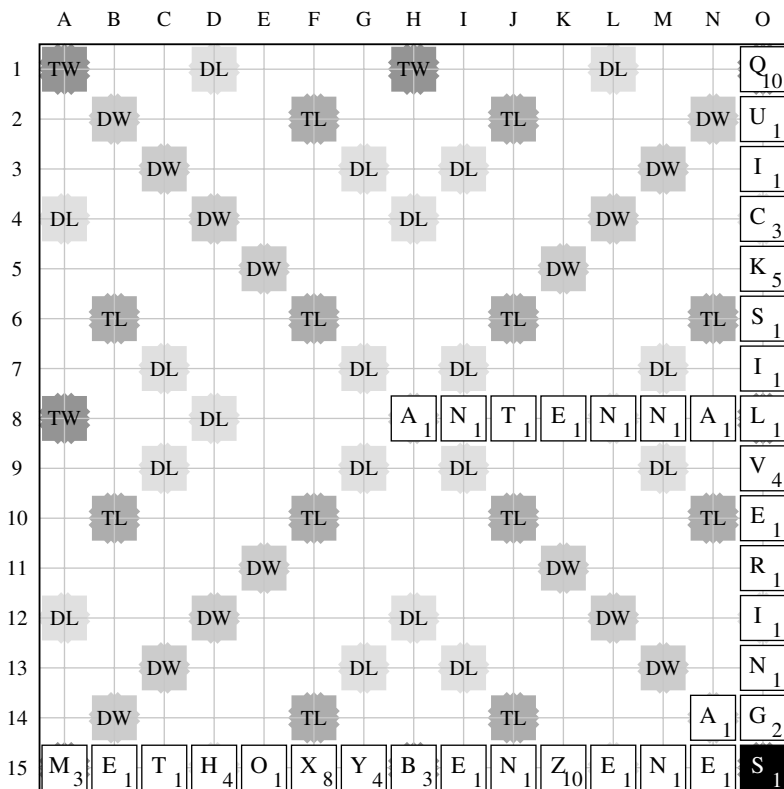
A ₁	A ₁	B ₃	B ₃	D ₂	D ₂	S ₁
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Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Word	Span	New tiles	Score
1	AMNESIA	H8–N8	7	70
2	TECH	O6–O9	4	63
3	TECHNIQUE	O6–O14	5	33
4	MICROTECHNIQUE	O1–O14	5	99
5	RE	N14–O14	1	4
6	ZINE	K15–N15	4	16
7	HYDROXYZINE	D15–N15	7	173
8	HYDROXYZINES	D15–O15	1	213

CSW24: $k = 1$, 225 points



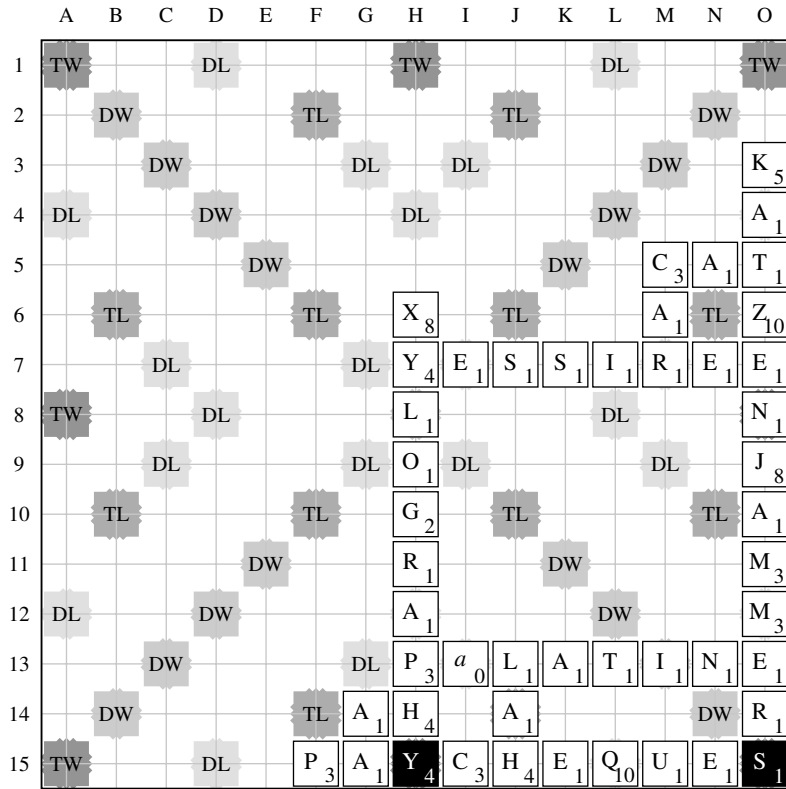
Rack A₁ B₃ C₃ D₂ D₂ D₂ S₁

Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Word	Span	New tiles	Score
1	ANTENNA	H8–N8	7	66
2	SILVER	O6–O11	6	51
3	SILVERING	O6–O14	3	14
4	QUICKSILVERING	O1–O14	5	108
5	AG	N14–O14	1	6
6	BENZENE	H15–N15	7	109
7	METHOXYBENZENE	A15–N15	7	182
8	METHOXYBENZENES	A15–O15	1	225

NWL23: $k = 2$, 459 points



Rack: E₁ E₁ E₁ E₁ E₁ S₁ Y₄

Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Word	Span	New tiles	Score
1	LOG	H8–H10	3	8
2	XYLOGRAPH	H6–H14	6	26
3	YESSIREE	H7–O7	7	63
4	CAR	M5–M7	2	5
5	CAT	M5–O5	2	5
6	KAT	O3–O5	2	8
7	PALATINE	H13–O13	7	68
8	JAMMER	O9–O14	5	20
9	KATZENJAMMER	O3–O14	2	108
10	LA	J13–J14	1	4
11	CHEQUE	I15–N15	6	36
12	AH	G14–H14	1	5
13	PA	F15–G15	2	6
14	PAYCHEQUES	F15–O15	2	459

CSW24: $k = 2$, 483 points

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	TW			DL				TW				DL			M ₃
2		DW				TL				TL				DW	E ₁
3			DW				DL	Q ₁₀	DL				DW		T ₁
4	DL			DW				U ₁				DW			H ₄
5					DW			A ₁			DW				O ₁
6		TL				TL		L ₁		TL				TL	X ₈
7			DL				DL	I ₁	DL				DL		Y ₄
8	TW			DL				F ₄	I ₁	R ₁	E ₁	B ₃	O ₁	M ₃	B ₃
9			DL				DL	I ₁	DL		A ₁		DL		E ₁
10		TL				TL		C ₃	A ₁	R ₁	T ₁			TL	N ₁
11					DW			A ₁			DW				Z ₁₀
12	DL			DW				T ₁				DW			E ₁
13			DW				DL	O ₁	DL				DW		N ₁
14		DW				TL	E ₁	R ₁	A ₁	TL				DW	E ₁
15	TW			DL		J ₈	A ₁	Y ₄	H ₄	A ₁	W ₄	K ₅	E ₁	R ₁	S

Rack

D ₂	D ₂	E ₁	E ₁	E ₁	S ₁	Y ₄
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Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Word	Span	New tiles	Score
1	ALIF	H5–H8	4	14
2	FIREBOMB	H8–O8	7	110
3	EAT	K8–K10	2	3
4	CART	H10–K10	3	8
5	CAT	H10–H12	2	6
6	QUALIFICATOR	H3–H14	5	27
7	BENZENE	O8–O14	6	19
8	METHOXYBENZENE	O1–O14	7	182
9	ERA	G14–I14	2	3
10	HAWKER	I15–N15	6	26
11	JA	F15–G15	2	11
12	JAYHAWKERS	F15–O15	2	483

NWL23: $k = 3, 882$ points

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	TW			DL				TW				DL			D ₂
2		DW				TL				TL				DW	E ₁
3			DW				DL		DL				DW		M ₃
4	DL			DW				O ₁				DW			I ₁
5					DW			B ₃			DW				S ₁
6		TL				TL		J ₈		TL				TL	E ₁
7			DL				DL	E ₁	DL				DL		M ₃
8	R ₁	A ₁	B ₃	I ₁	E ₁	T ₁	I ₁	C ₃				DL			I ₁
9	I ₁		DL				DL	T ₁	DL				DL		Q ₁₀
10	C ₃	TL				TL		I ₁		TL				TL	U ₁
11	K ₅				DW			V ₄	E ₁	N ₁	D ₂	E ₁	T ₁	T ₁	A ₁
12	S ₁			DW				I ₁				DW			V ₄
13	H ₄		DW				DL	Z ₁₀	DL				DW		E ₁
14	A ₁	DW				TL	A ₁	E ₁		TL				A ₁	R ₁
15	W ₄	H ₄	I ₁	P ₃	P ₀	E ₁	R ₁	S ₁	N ₁	A ₁	P ₀	P ₃	E ₁	R ₁	S ₁

Rack

F ₄	F ₄	G ₂	G ₂	S ₁	S ₁	W ₄
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Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Word	Span	New tiles	Score
1	OBJECT	H4–H9	6	36
2	OBJECTIVIZE	H4–H14	5	35
3	VENDETTA	H11–O11	7	74
4	QUAVER	O9–O14	5	22
5	SEMIQUAVER	O5–O14	4	72
6	DEMISEMIQUAVER	O1–O14	4	96
7	RABIETIC	A8–H8	7	89
8	RICKSHA	A8–A14	6	17
9	AE	G14–H14	1	2
10	HIPPER	B15–G15	6	15
11	AR	N14–O14	1	4
12	NAPPER	I15–N15	6	12
13	WHIPPERSNAPPERS	A15–O15	3	882

CSW24: $k = 3, 912$ points

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	TW			DL				M ₃				DL			TW
2		DW				TL		E ₁		TL				DW	Q ₁₀
3			DW				DL	T ₁	DL				DW		U ₁
4	DL			DW				H ₄				DW			A ₁
5					DW			O ₁	R ₁	I ₁	E ₁	N ₁	T ₁	A ₁	L ₁
6		TL				TL		X ₈		TL				TL	I ₁
7	K ₅		DL				DL	Y ₄	DL				DL		F ₄
8	A ₁			DL				B ₃				DL			I ₁
9	B ₃		DL				DL	E ₁	U ₁	T ₁	E ₁	C ₃	T ₁	I ₁	C ₃
10	E ₁	TL				TL		N ₁		TL				TL	A ₁
11	L ₁				DW			Z ₁₀			DW				T ₁
12	J ₈			DW				E ₁				DW			I ₁
13	O ₁		DW				DL	N ₁	DL				DW		V ₄
14	U ₁	S ₁				TL	E ₁	E ₁		TL				A ₁	E ₁
15	W ₄	H ₄	I ₁	P ₀	P ₀	E ₁	R ₁	S ₁	N ₁	A ₁	P ₃	P ₃	E ₁	R ₁	S ₁

Rack

E ₁	F ₄	G ₂	G ₂	S ₁	S ₁	W ₄
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Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Word	Span	New tiles	Score
1	BENZENE	H8–H14	7	88
2	METHOXYBENZENE	H1–H14	7	182
3	EE	G14–H14	1	2
4	HIPPER	B15–G15	6	9
5	SH	B14–B15	1	10
6	OU	A13–A14	2	4
7	KABELJOU	A7–A14	6	87
8	EUTECTIC	H9–O9	7	64
9	CAT	O9–O11	2	5
10	ORIENTAL	H5–O5	7	66
11	ALIF	O4–O7	3	8
12	QUALIFICATIVE	O2–O14	6	93
13	AE	N14–O14	1	4
14	NAPPER	I15–N15	6	15
15	WHIPPERSNAPPERS	A15–O15	3	912

NWL23: $k = 4$, 1130 points

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	TW		P ₃	DL				TW				DL			TW
2		DW	L ₁			TL				TL				DW	V ₄
3			E ₁				DL		DL				DW		E ₁
4	DL		T ₁	DW				DL				DW			N ₁
5			h ₀		DW							DW			T ₁
6	B ₃	A ₁	Y ₄			TL				TL				O ₁	R ₁
7	A ₁	D ₂	S ₁				DL		DL				DL	S ₁	I ₁
8	L ₁		M ₃	DL				F ₄	I ₁	R ₁	E ₁	R ₁	E ₁	E ₁	L ₁
9	D ₂	U ₁	O ₁				DL		DL	E ₁			DL		O ₁
10	A ₁	TL	G ₂			TL				F ₄				TL	Q ₁₀
11	C ₃	A ₁	R ₁		DW			J ₈		I ₁	DW				U ₁
12	H ₄		A ₁	DW				O ₁	S ₁	T ₁	I ₁	N ₁	A ₁	T ₁	I ₁
13	I ₁		P ₃	O ₁	E ₁	T ₁	I ₁	C ₃	DL				DW		Z ₁₀
14	N ₁	A ₁	h ₀		T ₁	TL		K ₅	A ₁	TL				DW	E ₁
15	O ₁	X ₈	Y ₄	H ₄	A ₁	E ₁	M ₃	O ₁	G ₂	L ₁	O ₁	B ₃	I ₁	N ₁	S ₁

Rack

N ₁	N ₁	O ₁	O ₁	R ₁	S ₁	Y ₄
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Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Word	Span	New tiles	Score
1	FIRER	H8–L8	5	18
2	FIREREEL	H8–O8	3	33
3	OSE	N6–N8	2	5
4	OR	N6–O6	1	2
5	REFIT	J8–J12	4	16
6	OSTINATI	H12–O12	7	70
7	QUIZ	O10–O13	3	22
8	VENTRILQUIZE	O2–O14	7	87
9	JOCK	H11–H14	3	17
10	POETIC	C13–H13	5	22
11	GRAPH	C10–C14	4	7
12	NAH	A14–C14	2	4
13	AX	B14–B15	1	9
14	KA	H14–I14	1	6
15	GLOBIN	I15–N15	6	15
16	ETA	E13–E15	2	3
17	HAEM	D15–G15	3	13
18	CHI	A11–A13	3	13
19	BALDACHIN	A6–A14	5	51
20	BAY	A6–C6	2	10
21	THY	C4–C6	2	5
22	PLETHYSMOGRAPH	C1–C14	6	48
23	ADS	A7–C7	1	7
24	AR	B11–C11	1	5
25	DUO	A9–C9	1	4
26	OXYHAEMOGLOBINS	A15–O15	4	1130

CSW24: $k = 4$, 1349 points

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	TW			DL				TW				DL			Q ₁₀
2	<i>h</i> ₀	DW				TL				TL				DW	U ₁
3	Y ₄		DW				DL		DL				DW		I ₁
4	P ₃			DW				DL				DW			C ₃
5	O ₁				DW						DW				K ₅
6	C ₃	TL				TL				TL				TL	S ₁
7	H ₄		DL				DL		DL				DL		I ₁
8	O ₁			DL				T ₁	U ₁	T ₁	O ₁	R ₁	I ₁	A ₁	L ₁
9	N ₁		DL				DL		DL	I ₁			DL		V ₄
10	D ₂	E ₁	F ₄	O ₁	R ₁	E ₁	S ₁	T ₁	E ₁	R ₁				TL	E ₁
11	R ₁				DW					E ₁	W ₄				R ₁
12	I ₁			DW				J ₈	U ₁	D ₂	O ₁	DW			I ₁
13	A ₁		DW				DL	A ₁	DL		O ₁		DW		N ₁
14	S ₁	<i>h</i> ₀				TL		M ₃	A ₁	TL	T ₁			A ₁	G ₂
15	M ₃	E ₁	T ₁	H ₄	O ₁	X ₈	Y ₄	B ₃	E ₁	N ₁	Z ₁₀	E ₁	N ₁	E ₁	S ₁

Rack **B₃** **E₁** **E₁** **E₁** **M₃** **S₁** **Z₁₀**

Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Word	Span	New tiles	Score
1	TUTOR	H8-L8	5	12
2	TUTORIAL	H8-O8	3	24
3	TIRED	J8-J12	4	8
4	JUDO	H12-K12	3	20
5	JAM	H12-H14	2	12
6	WOOT	K11-K14	3	24
7	SILVER	O6-O11	5	9
8	SILVERING	O6-O14	3	14
9	QUICKSILVERING	O1-O14	5	108
10	AG	N14-O14	1	6
11	ENE	L15-N15	3	6
12	MA	H14-I14	1	4
13	EN	I15-J15	2	4
14	FORESTER	C10-J10	7	63
15	DEFORESTER	A10-J10	2	16
16	HOND	A7-A10	3	24
17	CHONDRI	A6-A12	3	14
18	HYPOCHONDRIA	A2-A13	5	25
19	HYPOCHONDRIAS	A2-A14	1	23
20	SH	A14-B14	1	2
21	ETHOXY	B15-G15	6	24
22	METHOXYBENZENES	A15-O15	4	1349

NWL23: $k = 5$, 1401 points

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	H ₄			DL				TW				DL			TW
2	Y ₄	DW				TL				TL				DW	
3	P ₃		DW				DL		DL				DW		
4	E ₁			DW				DL				DW			DL
5	R ₁				O ₁						DW				
6	V ₄	TL			V ₄	TL				TL				TL	M ₃
7	E ₁		DL		E ₁		DL		DL				DL		Y ₄
8	N ₁	A ₁	P ₃	E ₁	R ₁	I ₁	E ₁	S ₁				DL			X ₈
9	T ₁		A ₁		F ₄		DL		DL				DL		A ₁
10	I ₁	TL	R ₁		U ₁	TL				TL		Q ₁₀		TL	M ₃
11	L ₁		A ₁		L ₁			J ₈			DW	U ₁			O ₁
12	A ₁		N ₁	DW	F ₄	A ₁	D ₂	O ₁				A ₁			E ₁
13	T ₁		O ₁		I ₁		DL	C ₃	I ₁	T ₁	E ₁	R ₁	DW		B ₃
14	E ₁	DW	I ₁		L ₁	TL	O ₁	K ₅		I ₁		T ₁	O ₁	E ₁	A ₁
15	D ₂	I ₁	C ₃	H ₄	L ₁	O ₁	R ₁	O ₁	B ₃	E ₁	N ₁	Z ₁₀	E ₁	N ₁	E ₁

Rack **D₂** **E₁** **G₂** **I₁** **L₁** **O₁** **Z₁₀**

Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Word	Span	New tiles	Score
1	APERIES	B8-H8	7	70
2	NAPERIES	A8-H8	1	30
3	VENT	A6-A9	3	7
4	VENTILATE	A6-A14	5	13
5	HYPERVENTILATE	A1-A14	5	78
6	OVER	E5-E8	3	14
7	OVERFULFIL	E5-E14	6	38
8	FADO	E12-H12	3	9
9	JOCK	H11-H14	3	17
10	OR	G14-G15	2	8
11	OR	F15-G15	1	2
12	PARANOIC	C8-C15	7	76
13	ICH	B15-D15	2	12
14	CITER	H13-L13	4	8
15	TIE	J13-J15	2	5
16	BEN	I15-K15	2	5
17	QUART	L10-L14	4	28
18	TOEA	L14-O14	3	8
19	EN	M15-N15	2	6
20	AMOEBA	O9-O14	5	11
21	MYXAMOEBA	O6-O14	3	75
22	DICHLOROBENZENE	A15-O15	5	1401

CSW24: $k = 5$, 1478 points

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	TW			DL				TW				DL			Q ₀
2	<i>h</i> ₀	DW				TL				TL				DW	U ₁
3	Y ₄		DW				DL		DL				DW		I ₁
4	P ₃			DW				DL				DW			C ₃
5	O ₁				DW							DW			K ₅
6	C ₃	L ₁	A ₁	P ₃		TL				TL				TL	S ₁
7	H ₄		DL	E ₁			DL		DL				DL		I ₁
8	O ₁			N ₁				T ₁	U ₁	T ₁	O ₁	R ₁	I ₁	A ₁	L ₁
9	N ₁		DL	N ₁			DL		DL	I ₁			DL		V ₄
10	D ₂	TL		y ₀	I ₁	E ₁	L ₁	D ₂	E ₁	R ₁				TL	E ₁
11	R ₁			W ₄	DW					E ₁	W ₄				R ₁
12	I ₁			O ₁				J ₈	U ₁	D ₂	O ₁	DW			I ₁
13	A ₁		DW	R ₁			DL	A ₁	DL		O ₁		DW		N ₁
14	S ₁	O ₁			T ₁	I ₁	TL		M ₃	A ₁	TL	T ₁		A ₁	G ₂
15	M ₃	E ₁	T ₁	H ₄	O ₁	X ₈	Y ₄	B ₃	E ₁	N ₁	Z ₁₀	E ₁	N ₁	E ₁	S ₁

Rack **B₃** **E₁** **F₄** **H₄** **M₃** **S₁** **Z₁₀**

Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Word	Span	New tiles	Score
1	TUTOR	H8-L8	5	12
2	TUTORIAL	H8-O8	3	24
3	TIRED	J8-J12	4	8
4	JUDO	H12-K12	3	20
5	JAM	H12-H14	2	12
6	WOOT	K11-K14	3	24
7	SILVER	O6-O11	5	9
8	SILVERING	O6-O14	3	14
9	QUICKSILVERING	O1-O14	5	108
10	AG	N14-O14	1	6
11	ENE	L15-N15	3	6
12	MA	H14-I14	1	4
13	EN	I15-J15	2	4
14	YIELDER	D10-J10	6	9
15	PENNY	D6-D10	4	7
16	PENNYWORT	D6-D14	4	26
17	CLAP	A6-D6	3	10
18	CHONDRI	A6-A12	6	42
19	HYPOCHONDRIA	A2-A13	5	25
20	HYPOCHONDRIAS	A2-A14	1	23
21	SO	A14-B14	1	4
22	ET	B15-C15	2	4
23	TI	D14-E14	1	2
24	OXY	E15-G15	3	15
25	METHOXYBENZENES	A15-O15	5	1478

NWL23: $k = 6$, 1723 points

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	O ₁	X ₈	Y ₄	P ₃	H ₄	E ₁	N ₁	B ₃	U ₁	T ₁	A ₁	Z ₁₀	O ₁	N ₁	E ₁
2	P ₃	E ₁	E ₁	R ₁		TL	E ₁	L ₁		TL	W ₄	O ₁		U ₁	S ₁
3	A ₁	D ₂	DW	E ₁			DL	A ₁	DL			O ₁	DW		T ₁
4	C ₃			Q ₁₀				D ₂				G ₂			A ₁
5	I ₁			U ₁	DW			D ₂			DW	A ₁			B ₃
6	F ₄	TL		A ₁	D ₂	O ₁	R ₁	E ₁		TL		M ₃		TL	L ₁
7	I ₁		DL	L ₁			DL	R ₁	DL			E ₁	DL		I ₁
8	C ₃			I ₁				L ₁	E ₁	G ₂	I ₁	T ₁			S ₁
9	A ₁		DL	F ₄			DL	I ₁	DL			E ₁	A ₁	c ₀	H ₄
10	T ₁	TL		Y ₄		TL		K ₅		TL				TL	M ₃
11	I ₁			I ₁	DW			E ₁			DW				E ₁
12	O ₁			N ₁				DL				DW			N ₁
13	N ₁		DW	G ₂			DL		DL				DW		T ₁
14	S ₁	DW				TL				TL				DW	S ₁
15	TW			DL				TW				DL			TW

Rack **B₃ E₁ O₁ P₃ R₁ X₈ Z₁₀**

Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Word	Span	New tiles	Score
1	LIKE	H8-H11	4	16
2	LADDERLIKE	H2-H11	6	18
3	LEGIT	H8-L8	4	7
4	OOGAMETE	L2-L9	7	72
5	EACH	L9-O9	3	7
6	STABLISH	O2-O9	7	92
7	STABLISHMENT	O2-O13	4	20
8	STABLISHMENTS	O2-O14	1	20
9	ADORE	D6-H6	4	8
10	QUALIFY	D4-D10	6	46
11	REQUALIFYING	D2-D13	5	56
12	PEER	A2-D2	3	12
13	PACIFIC	A2-A8	6	57
14	PACIFICATIONS	A2-A14	6	23
15	ED	B2-B3	1	6
16	YE	C1-C2	1	5
17	NE	G1-G2	2	4
18	HEN	E1-G1	2	6
19	AW	K1-K2	2	10
20	UTA	I1-K1	2	3
21	NU	N1-N2	2	8
22	ON	M1-N1	1	2
23	OXYPHENBUTAZONE	A1-O1	6	1723

CSW24: $k = 6$, 1721 points

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	O ₁	X ₈	Y ₄	P ₃	H ₄	E ₁	N ₁	B ₃	U ₁	T ₁	A ₁	Z ₁₀	O ₁	N ₁	E ₁
2	P ₃	E ₁	A ₁	R ₁		D ₂	O ₁	R ₁	R ₁	TL		O ₁	B ₃	DW	J ₈
3	A ₁	D ₂	R ₁	E ₁	D ₂		DL	A ₁	E ₁			O ₁	DW		A ₁
4	C ₃		A ₁	Q ₀	U ₁	I ₁	L ₁	I ₁	n ₀	E ₁		G ₂			C ₃
5	I ₁		K ₅	U ₁	E ₁		I ₁	N ₁	T ₁	R ₁	E ₁	A ₁	T ₁		U ₁
6	F ₄	TL	S ₁	a ₀	L ₁	TL		W ₄		TL		M ₃		TL	L ₁
7	Y ₄		DL	L ₁			DL	A ₁	DL			E ₁	DL		A ₁
8	I ₁			I ₁				S ₁				T ₁	O ₁	O ₁	T ₁
9	N ₁		DL	F ₄			DL	H ₄	DL			E ₁	DL		I ₁
10	G ₂	TL		I ₁		TL		I ₁		TL		S ₁		TL	O ₁
11				E ₁	DW			N ₁			DW				N ₁
12	DL			D ₂				G ₂				DW			S ₁
13			DW				DL		DL				DW		
14		DW				TL				TL				DW	
15	TW			DL				TW				DL			TW

Rack **B₃** **E₁** **E₁** **O₁** **P₃** **Y₄** **Z₁₀**

Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Player	Word	Span	New tiles	Score
1	A	SH	H8-H9	2	10
2	A	RAINWASH	H2-H9	6	15
3	A	INTREAT	G5-M5	6	14
4	A	AM	L5-L6	1	4
5	A	AE	H3-I3	1	3
6	A	DORR	F2-I2	3	11
7	A	AQUILINE	C4-J4	7	89
8	A	DUE	E3-E5	2	8
9	A	ARAK	C2-C5	3	16
10	A	EQUAL	D3-D7	4	24
11	A	ADRED	A3-E3	2	7
12	A	PA	A2-A3	1	4
13	A	OOGAMETES	L2-L10	7	76
14	A	OB	L2-M2	1	4
15	A	XED	B1-B3	2	32
16	A	TOO	L8-N8	2	3
17	A	SAL	C6-E6	2	16
18	A	ON	M1-N1	2	6
19	A	RAINWASHING	H2-H12	3	20
20	A	UTA	I1-K1	3	7
21	A	HEN	E1-G1	3	11
22	A	LATI	O6-O9	4	24
23	A	REQUALIFIED	D2-D12	6	54
24	A	JACULATIONS	O2-O12	7	74
25	B	PACIFYING	A2-A10	7	119
26	A	OXYPHENBUTAZONE	A1-O1	6	1721

NWL23: $k = 7$, 1786 points

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	O ₁	X ₈	Y ₄	P ₃	H ₄	E ₁	N ₁	B ₃	U ₁	T ₁	A ₁	Z ₁₀	O ₁	N ₁	E ₁
2	P ₃	E ₁	A ₁	R ₁		h ₀	A ₁	L ₁	T ₁	TL	W ₄	O ₁		O ₁	S ₁
3	A ₁	D ₂	DW	E ₁			R ₁	A ₁	s ₀			O ₁	DW		T ₁
4	C ₃			Q ₀		U ₁	R ₁	D ₂				G ₂			A ₁
5	I ₁			U ₁	DW		O ₁	D ₂			DW	A ₁			B ₃
6	F ₄	TL		A ₁	V ₄	O ₁	W ₄	E ₁	R ₁	TL		M ₃	E ₁	R ₁	L ₁
7	I ₁		DL	L ₁			I ₁	R ₁	E ₁			E ₁	DL		I ₁
8	C ₃			I ₁		U ₁	N ₁	L ₁	E ₁	D ₂		T ₁			S ₁
9	A ₁		DL	F ₄			G ₂	I ₁	DL			E ₁	DL		H ₄
10	T ₁	TL		Y ₄		TL		K ₅		TL				TL	M ₃
11	I ₁			I ₁	DW			E ₁			DW				E ₁
12	O ₁			N ₁				DL				DW			N ₁
13	N ₁		DW	G ₂			DL		DL				DW		T ₁
14	S ₁	DW				TL				TL				DW	S ₁
15	TW			DL				TW				DL			TW

Rack **B₃ E₁ N₁ O₁ P₃ X₈ Z₁₀**

Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Player	Word	Span	New tiles	Score
1	A	UNLED	F8–J8	5	12
2	A	ARROWING	G2–G9	7	66
3	A	HALT	F2–I2	3	3
4	A	UTS	I1–I3	2	2
5	A	UTA	I1–K1	2	3
6	A	WO	K2–L2	2	10
7	A	OOGAMETE	L2–L9	7	74
8	A	MERL	L6–O6	3	8
9	A	STABLISH	O2–O9	7	92
10	A	OS	N2–O2	1	4
11	A	REE	I6–I8	2	4
12	A	VOW	E6–G6	2	11
13	A	AL	D6–D7	2	12
14	A	REQUALIFY	D2–D10	7	100
15	A	YA	C1–C2	2	7
16	A	ED	B2–B3	2	12
17	A	URD	F4–H4	2	6
18	A	ON	M1–N1	2	4
19	A	HE	E1–F1	2	6
20	A	REQUALIFYING	D2–D13	3	56
21	A	STABLISHMENTS	O2–O14	5	21
22	A	PACIFIC	A2–A8	7	116
23	A	PACIFICATIONS	A2–A14	6	23
24	B	LADDERLIKE	H2–H11	7	89
25	A	OXYPHENBUTAZONE	A1–O1	7	1786

Construction due to Bob Lucassen (2024); the table is normalized from the independently verified rack certificate.

CSW24: $k = 7$, 1787 points

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	O ₁	X ₈	Y ₄	P ₃	H ₄	E ₁	N ₁	B ₃	U ₁	T ₁	A ₁	Z ₁₀	O ₁	N ₁	E ₁
2	P ₃	E ₁	a ₀	R ₁		R ₁	O ₁	R ₁	T ₁	TL	M ₃	O ₁	O ₁	DW	J ₈
3	A ₁	D ₂	D ₂	E ₁	E ₁	m ₀	DL	A ₁	DL	R ₁	E ₁	O ₁	DW		A ₁
4	C ₃			Q ₀			D ₂	I ₁	V ₄	U ₁	L ₁	G ₂	E ₁		C ₃
5	I ₁			U ₁	DW			N ₁		B ₃	I ₁	A ₁	S ₁		U ₁
6	F ₄	TL		A ₁		TL		W ₄		TL	O ₁	M ₃		TL	L ₁
7	Y ₄		DL	L ₁			DL	A ₁	DL		R ₁	E ₁	DL		A ₁
8	I ₁			I ₁				S ₁			A ₁	T ₁			T ₁
9	N ₁		DL	F ₄			DL	H ₄	DL		T ₁	E ₁	T ₁	R ₁	I ₁
10	G ₂	TL		I ₁		TL		I ₁		TL	I ₁	S ₁		TL	O ₁
11				E ₁	DW			N ₁			V ₄				N ₁
12	DL			D ₂				G ₂			E ₁	DW			S ₁
13			DW				DL		DL				DW		
14		DW				TL				TL				DW	
15	TW			DL				TW				DL			TW

Rack **A₁ B₃ E₁ O₁ P₃ X₈ Z₁₀**

Black tiles are newly played; italic letters are blanks; the final rack appears below the board.

Complete tile-placement history

No.	Player	Word	Span	New tiles	Score
1	A	SH	H8-H9	2	10
2	A	RAINWASH	H2-H9	6	15
3	A	DIVULGE	G4-M4	6	24
4	A	ES	M4-M5	1	2
5	A	GAM	L4-L6	2	8
6	A	RE	K7-L7	2	9
7	A	OOGAMETE	L2-L9	4	12
8	A	OO	L2-M2	1	2
9	A	RUB	J3-J5	2	5
10	A	ORT	G2-I2	2	3
11	A	ER	F1-F2	2	10
12	A	DEEM	C3-F3	4	10
13	A	YAD	C1-C3	2	6
14	A	IS	K10-L10	2	14
15	A	ADDEEM	A3-F3	2	7
16	A	PEA	A2-C2	2	18
17	A	ON	M1-N1	2	4
18	A	UT	I1-J1	2	4
19	A	HEN	E1-G1	2	8
20	A	TETRI	K9-O9	4	8
21	A	IONS	O9-O12	3	5
22	A	RAINWASHING	H2-H12	3	20
23	A	EQUAL	D3-D7	4	28
24	A	REQUALIFIED	D2-D12	6	55
25	A	JACULATIONS	O2-O12	7	119
26	A	MELIORATIVE	K2-K12	7	140
27	B	PACIFYING	A2-A10	7	119
28	A	OXYPHENBUTAZONE	A1-O1	7	1787